AI MITRA

AI Dictionary

A friendly guide to Artificial Intelligence for Classes 8–10

Explore AI, one simple word at a time.
Understand important terms and connect them to familiar situations from everyday life.

Simple English • Clear explanations • Real-life examples

AI Mitra | Explore AI Together

How to use this dictionary

Read each explanation and connect it to the real-life example. You do not need to memorise every word at once—learn a few terms at a time and notice where you see them in everyday technology.

Remember: AI tools can make mistakes. For schoolwork, compare important information with your textbook, teacher or trusted sources. Never share passwords, home addresses, phone numbers or private photographs with AI tools.

What you will find

1. AI Foundations

Start with the core ideas behind Artificial Intelligence and how AI systems learn from information.

2. Generative AI & Language

Explore the technologies behind chat assistants, generated writing, images, speech and other creative content.

1. AI Foundations

Start with the core ideas behind Artificial Intelligence and how AI systems learn from information.

Artificial Intelligence (AI)

Explanation: Artificial intelligence is technology that helps computers perform tasks that usually need human abilities, such as recognising pictures, understanding language or making suggestions. It can support people, but it does not think exactly like a human.

Real-life example: A phone recognises your face and unlocks the screen.

Machine Learning (ML)

Explanation: Machine learning is a way for computers to learn patterns from examples instead of being given every rule by a person. The patterns help the computer make predictions or decisions on new information.

Real-life example: A music app notices the songs you enjoy and suggests similar songs.

Deep Learning

Explanation: Deep learning is a type of machine learning that uses many connected layers to find complex patterns in large amounts of data. It is often used for tasks involving images, speech and language.

Real-life example: A photo app can learn to recognise different animals in pictures.

Algorithm

Explanation: An algorithm is a clear set of steps used to solve a problem or complete a task. Computers follow algorithms to process information and produce results.

Real-life example: A recipe gives you the steps to make a sandwich in the right order.

AI Model

Explanation: An AI model is a computer system trained using data to recognise patterns or produce predictions and responses. Different models are designed for different tasks.

Real-life example: A trained model looks at a picture and predicts whether it shows a cat or a dog.

Data

Explanation: Data is information that a computer can use, such as words, numbers, pictures, sounds or measurements. AI systems use data to learn patterns and perform tasks.

Real-life example: A class attendance list containing students' names and attendance is data.

Dataset

Explanation: A dataset is an organised collection of related information. It may contain text, numbers, images or audio and can be used to train or test an AI model.

Real-life example: A folder containing hundreds of labelled photos of cats and dogs is a dataset.

Training

Explanation: Training is the process in which an AI model studies examples and adjusts how it works so it can perform a task better. The quality and variety of examples can affect what the model learns.

Real-life example: Just as practising maths helps you improve, showing an AI many labelled examples helps it learn a task.

Prediction

Explanation: A prediction is an estimate about what may happen or what something might be, based on patterns in available information. Predictions can be useful, but they are not guaranteed to be correct.

Real-life example: A weather app predicts whether it may rain tomorrow using weather information.

Inference

Explanation: Inference happens when a trained AI model uses what it has learned to respond to new information. It applies learned patterns rather than repeating its training process.

Real-life example: After learning from many animal pictures, a model identifies an animal in a new photo.

2. Generative AI & Language

Explore the AI technologies behind chat assistants, generated writing, images, speech and other creative content.

Generative AI

Explanation: Generative AI creates new content, such as text, pictures, music or code, using patterns learned from examples. The result depends on the model, the information it has learned and the instructions it receives.

Real-life example: You describe a birthday card, and an AI tool creates a new design idea.

Large Language Model (LLM)

Explanation: A large language model is an AI model trained on very large amounts of text. It can work with language to answer questions, summarise information, explain topics and help with writing.

Real-life example: A chatbot explains a science lesson in simpler words when you ask it.

Chatbot

Explanation: A chatbot is a computer program that communicates with people through text or voice. AI chatbots can understand many everyday questions and generate replies, but their answers may need checking.

Real-life example: A school website chatbot answers questions about admission dates and school timings.

Prompt

Explanation: A prompt is the question, instruction or information you give an AI tool to guide its response. A clear prompt explains what you need and can help the tool give a more useful answer.

Real-life example: “Explain gravity using an example from a playground” is a prompt.

Prompt Engineering

Explanation: Prompt engineering means writing and improving instructions so an AI tool better understands what you need. It can include adding context, examples, limits and a preferred answer format.

Real-life example: Adding “I am in Class 8; use simple words and give two examples” improves a study prompt.

Token

Explanation: A token is a small piece of text that an AI language model processes. A token may be a whole word, part of a word, a number or punctuation, depending on how the text is divided.

Real-life example: Before a chatbot processes a long sentence, it breaks the text into smaller pieces called tokens.

Context Window

Explanation: A context window is the amount of information an AI model can consider at one time while responding. If a conversation or document is too long, some information may not fit into that window.

Real-life example: Think of a desk: only the books and notes placed on it can be used at that moment.

Multimodal AI

Explanation: Multimodal AI can work with more than one type of information, such as text, images, audio or video. It can combine these inputs to respond in different ways.

Real-life example: You show an AI a picture of a plant and ask it to describe what is visible.

Image Generation

Explanation: Image generation is the creation of a new picture by an AI system, often from a written description called a text prompt. The generated picture may not always match every detail you imagined.

Real-life example: You type “a treehouse beside a river at sunset” and an AI tool creates an illustration.

Speech Recognition

Explanation: Speech recognition converts spoken words into text that a computer can understand or work with. It is commonly used in voice typing, captions and voice assistants.

Real-life example: Your phone turns the words you speak into a written message.

Text-to-Speech (TTS)

Explanation: Text-to-speech turns written text into spoken audio. It can help people listen to articles, books, messages and instructions.

Real-life example: A phone reads a webpage aloud while you listen.

Diffusion Model

Explanation: A diffusion model is a type of generative AI that learns to create images by gradually changing random visual noise into a meaningful picture. It uses patterns learned during training to guide this process.

Real-life example: Imagine slowly turning a blurry, noisy picture into a clear drawing of a house.